142933: biotite–hypersthene–clinopyroxene–andesine mafic granulite, Meerawana Bore

Location and sampling

GLENBURGH (SG 50–6) AMG Zone 50, 421290E 7189830N

Sampled on 20 June 1998

The sample was taken from a partly buried, 1 m diameter boulder within a low rocky area on a slight south-dipping rise, about 350 m south of Meerawana Bore.

Tectonic unit/relations

This sample is from a mafic diorite that has been deformed, metamorphosed and intruded by pegmatite, and by granodiorite of the Nardoo Granite.

Petrographic description

The major minerals present in this sample are plagioclase (60 vol.%), clinopyroxene (18 vol.%), biotite (15 vol.%) and orthopyroxene (5 vol.%), with accessory amounts of amphibole, apatite and opaques (magnetite). The plagioclase is calcic andesine (about An_{38}), which is generally homogeneous. It forms irregular grains with some 120° triple-point grain junctions. Hypersthene forms irregular, elongate grains and in a few cases is at the core of grains of clinopyroxene. Clinopyroxene mainly forms irregular equant grains, in a few places enclosing hypersthene. Biotite occurs both as coarse, equant anhedral grains and as very thin, euhedral plates with regular sides. One minute grain with a high negative relief against plagioclase may be K-feldspar, but otherwise K-feldspar was not found. Magnetite forms coarse, very irregular grains possibly approaching 1 vol.% of the rock. Apatite is abundant. Zircon was not identified but irregular grains could be missed amongst the abundant fine grains of pyroxene. Fibrous materials, probably colourless amphibole, occur as pseudomorphs of complete mafic grains and as alteration in parts of other mafic grains. Some of this material seems to have parallel extinction, suggesting that it may be gedrite, or perhaps anthophyllite. Some of the fibrous material appears to be clinoamphibole. Biotite, clinopyroxene and hypersthene all contain abundant lattice-oriented inclusions. There is a thin scattering of inclusions in plagioclase at a size barely visible at higher magnification, but no recognizable saussuritization. The rock is generally coarse grained though a few grains of all minerals extend down to fine grain size. Apart from some biotite and less orthopyroxene, the principal minerals are equant. Most plagioclase is equant with simple grain boundaries, commonly straight or gently curved and commonly meeting in 120° triple points. Although biotite is oriented in local patches, there seems to be little overall foliation of the rock. The lack of euhedral forms, especially among the plagioclase grains, gives the rock a metamorphic appearance. The orthopyroxene is hypersthene, which is typical of granulite facies metamorphism, rather than enstatite, which is the more common igneous orthopyroxene. In an otherwise gabbroic rock such as this, unaltered plagioclase might be expected to be at least as calcic as labradorite. Here it is andesine. The present composition of the rock is not that of a typical gabbro. The anorthite content of the plagioclase is too low and the abundant biotite indicates more potassium than is present in most gabbroic rocks. The composition is that of a diorite, but it seems likely that the precursor was a gabbro that has been metasomatically altered. There is no textural evidence of metasomatism but

Table 22. Ion microprobe analytical results for sample 142933: biotite–hypersthene–clinopyroxene–andesine mafic granulite, Meerawana Bore

Grain .spot	*U (ppm)*	*Th (ppm)*	*Pb (ppm)*	*f206%*	$^{207}Pb/^{206}Pb$	$\pm 1\sigma$	$^{208}Pb/^{206}Pb$	$\pm 1\sigma$	$^{206}Pb/^{238}U$	$\pm 1\sigma$	$^{207}Pb/^{235}U$	$\pm 1\sigma$	*% concordance*	$^{207}Pb/^{206}Pb$ *Age*	$\pm 1\sigma$
1.1	528	481	219	0.118	0.12165	0.00055	0.25458	0.00090	0.3483	0.0076	5.841	0.133	97	1981	8
2.1	477	135	171	0.094	0.12204	0.00058	0.08089	0.00063	0.3462	0.0076	5.826	0.133	96	1986	9
3.1	1112	1705	513	0.049	0.12280	0.00039	0.44127	0.00081	0.3416	0.0074	5.785	0.129	95	1997	6
4.1	1788	3298	881	0.036	0.12329	0.00029	0.52486	0.00065	0.3463	0.0075	5.886	0.130	96	2004	4
5.1	923	1112	429	0.069	0.12140	0.00037	0.34245	0.00068	0.3676	0.0069	6.154	0.120	102	1977	5
6.1	242	181	103	0.260	0.12110	0.00084	0.21539	0.00136	0.3679	0.0070	6.143	0.130	102	1972	12
7.1	144	76	57	0.225	0.12125	0.00111	0.15511	0.00183	0.3582	0.0069	5.989	0.133	100	1975	16
8.1	115	85	48	0.336	0.12125	0.00131	0.20870	0.00239	0.3641	0.0070	6.088	0.142	101	1975	19
9.1	587	203	222	0.058	0.12315	0.00045	0.09414	0.00047	0.3606	0.0068	6.123	0.120	99	2002	6
10.1	364	368	158	0.181	0.12194	0.00063	0.28843	0.00111	0.3551	0.0067	5.970	0.121	99	1985	9
11.1	453	501	204	0.083	0.12249	0.00052	0.31665	0.00092	0.3630	0.0069	6.130	0.122	100	1993	8
12.1	651	813	298	0.057	0.12263	0.00042	0.35761	0.00078	0.3579	0.0068	6.051	0.119	99	1995	6
13.1	389	173	146	0.159	0.12200	0.00064	0.11842	0.00086	0.3491	0.0066	5.872	0.119	97	1986	9
14.1	1924	4396	1055	0.032	0.12171	0.00025	0.64799	0.00064	0.3589	0.0068	6.023	0.116	100	1981	4
15.1	306	335	137	0.148	0.12188	0.00068	0.31177	0.00123	0.3626	0.0069	6.094	0.124	101	1984	10
16.1	266	217	113	0.126	0.12228	0.00073	0.23005	0.00115	0.3616	0.0069	6.096	0.126	100	1990	11
17.1	634	909	299	0.111	0.12202	0.00046	0.41404	0.00092	0.3544	0.0067	5.963	0.117	98	1986	7
18.1	631	881	310	0.043	0.12159	0.00045	0.39364	0.00087	0.3756	0.0071	6.297	0.124	104	1980	7
19.1	895	1250	433	0.024	0.12238	0.00037	0.40437	0.00072	0.3677	0.0069	6.205	0.121	101	1991	5
20.1	1026	440	402	0.092	0.12309	0.00037	0.12573	0.00045	0.3633	0.0068	6.165	0.120	100	2002	5
21.1	738	1043	349	0.050	0.12275	0.00042	0.40113	0.00081	0.3595	0.0068	6.084	0.119	99	1997	6
22.1	658	915	309	0.112	0.12197	0.00045	0.40035	0.00090	0.3570	0.0067	6.003	0.118	99	1985	7
23.1	707	901	341	0.038	0.12220	0.00043	0.37027	0.00080	0.3748	0.0071	6.314	0.124	103	1989	6
24.1	1252	2012	612	0.045	0.12277	0.00031	0.45961	0.00066	0.3579	0.0067	6.057	0.117	99	1997	5
25.1	1251	2242	642	0.017	0.12189	0.00031	0.50608	0.00068	0.3657	0.0069	6.146	0.119	101	1984	4
26.1	933	1286	439	0.032	0.12195	0.00035	0.39194	0.00067	0.3606	0.0068	6.063	0.118	100	1985	5
27.1	1009	1545	486	0.035	0.12342	0.00037	0.42150	0.00074	0.3611	0.0068	6.145	0.120	99	2006	5
28.1	383	107	141	0.203	0.12200	0.00059	0.07928	0.00073	0.3539	0.0067	5.953	0.120	98	1986	9
29.1	311	278	130	0.190	0.11995	0.00071	0.25448	0.00121	0.3523	0.0067	5.826	0.120	99	1955	11
30.1	2010	3962	1006	0.047	0.12256	0.00026	0.55784	0.00060	0.3452	0.0065	5.832	0.112	96	1994	4

this might be expected to have been destroyed in the course of metamorphism at such high grade. The rock is a biotite–hypersthene–clinopyroxene–(calcic) andesine mafic granulite, in the hornblende–orthopyroxene–plagioclase subfacies of the granulite facies, probably after a metasomatically altered gabbroic precursor.

Zircon morphology

The zircons isolated from this sample are typically pale green-yellow, generally between 50 × 200 µm and 100 x 450 µm and commonly elongate and euhedral but with irregular, broken terminations. Many crystals have cylindrical cores that appear to be hollow or filled with fluid or melt inclusions. A minority are more equant and irregular in shape, and are generally structureless.

Analytical details

This sample was analysed on 8 and 13 August 1998. The counter deadtime during both analysis sessions was 32 ns. Ten analyses of the CZ3 standard obtained during the first analysis session indicated a Pb^*/U calibration error of 2.18 (1σ%). Analyses 1.1 to 4.1 were obtained during the first analysis session. During the second analysis session, ten analyses of the CZ3 standard indicated a Pb^*/U calibration error of 1.88 (1σ%). Common-Pb corrections were applied using Broken Hill common-Pb isotopic compositions for all analyses.

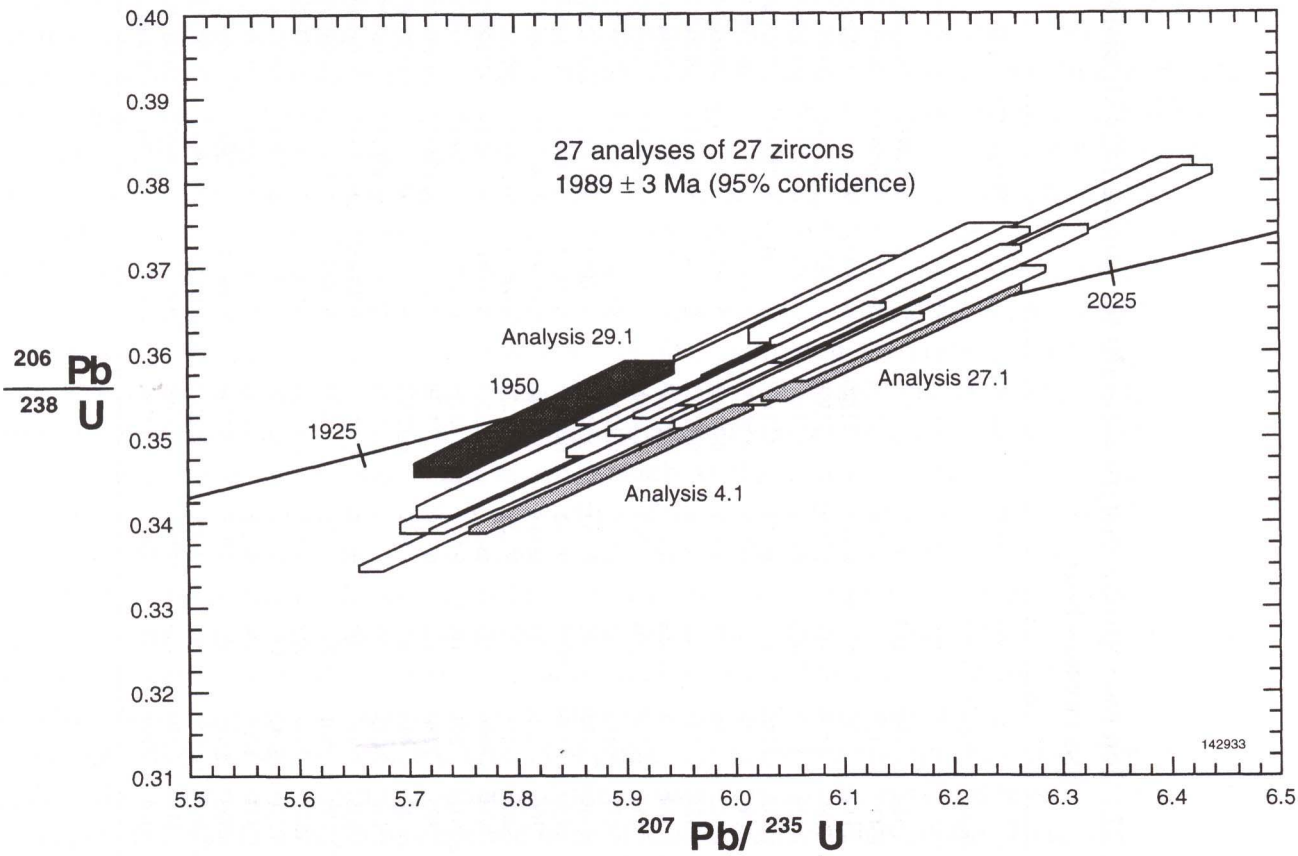

Figure 23. Concordia plot for sample 142933: biotite–hypersthene–clinopyroxene–andesine mafic granulite, Meerawana Bore

Results

Thirty analyses were obtained from 30 zircons. Results are given in Table 22 and shown on a concordia plot in Figure 23.

Interpretation

Twenty-seven concordant analyses have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean date of 1989 ± 3 Ma (chi-squared = 1.43). Concordant analysis 27.1, on a large (160 × 250 µm), irregular-shaped, structureless grain, and slightly discordant analysis 4.1, on the structureless rim of a 50 × 110 µm fragment having a hollow, inclusion-filled tubular core, have $^{207}Pb/^{206}Pb$ ratios defining a single population and indicating a weighted mean date of 2005 ± 3 Ma. Analysis 29.1 was obtained on a large (100 × 400 µ) elongate fragment having broken terminations and indicated a slightly lower $^{207}Pb/^{206}Pb$ ratio than that of the main population.

The date of 1989 ± 3 Ma indicated by twenty-seven concordant analyses of 27 zircons is interpreted as the time of igneous crystallization of the gabbroic or tonalitic precursor to the mafic gneiss. The date of 2005 ± 3 Ma indicated by analyses 27.1 and 4.1 may be of xenocryst zircons. Analysis 29.1 indicates a slightly younger $^{207}Pb/^{206}Pb$ date and is interpreted to be of an analysis site that has lost some radiogenic Pb during an ancient disturbance event.

STRATIGRAPHIC REFERENCE:

OCCHIPINTI, S. A., and SHEPPARD, S., 2001, Geology of the Glenburgh 1:100 000 sheet: Western Australia Geological Survey, 1:100 000 Geological Series Explanatory Notes, 37p.

Recommended reference for this publication:

NELSON, D. R., 1999, 142933: bio-hypersthene-clinopyroxene-andesine mafic granulite, Meerawana Bore; in Compilation of geochronology data, 1998: Western Australia Geological Survey, Record 1999/2, p. 90–93.
OR
NELSON, D. R., 1999, 142933: bio-hypersthene-clinopyroxene-andesine mafic granulite, Meerawana Bore; Geochronology dataset 320; in Compilation of geochronology data, June 2006 update: Western Australia Geological Survey.

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